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# STUDY OF NETWORK CABLES AND PREPARATION OF STRAIGHT-THROUGH AND CROSSOVER CABLES

## AIM:

To study the different types of network cables and prepare straight-through and crossover Ethernet cables using a crimping tool and RJ45 connectors.

## ALGORITHM:

1. Study the different types of network cables such as UTP, STP, coaxial and fibre optic cables.
2. Collect the required materials such as Ethernet cable, RJ45 connectors and crimping tool.
3. Cut the Ethernet cable to the required length.
4. Strip approximately 1.5 cm of the outer insulation from both ends of the cable.
5. Untwist and arrange the eight wires according to the required colour coding.
6. For a straight-through cable, arrange both ends using the same wiring standard (T568A or T568B).
7. For a crossover cable, arrange one end using T568A and the other end using T568B.
8. Trim the wires evenly and insert them into the RJ45 connectors.
9. Use the crimping tool to secure the connectors at both ends.
10. Test the prepared cables by connecting suitable network devices and checking connectivity.

## TYPES OF NETWORK CABLES:

### 1. Unshielded Twisted Pair (UTP) Cable

UTP cables consist of twisted pairs of copper wires without additional shielding. They are inexpensive, easy to install and commonly used in Ethernet networks.

### 2. Shielded Twisted Pair (STP) Cable

STP cables contain shielding to reduce electromagnetic interference and noise. They are used in environments where electrical interference is high.

### 3. Coaxial Cable

Coaxial cables consist of a central conductor surrounded by insulation and shielding. They are commonly used in television networks and broadband connections.

### 4. Fibre Optic Cable

Fibre optic cables transmit data using light signals. They provide high bandwidth, high transmission speed and support long-distance communication.

## PROCEDURE:

### A) STRAIGHT-THROUGH CABLE

1. Take an Ethernet cable and remove approximately 1.5 cm of insulation from both ends.
2. Arrange the wires at both ends according to the T568B colour code.
3. The colour sequence is:
    1. White-Orange
    2. Orange
    3. White-Green
    4. Blue
    5. White-Blue
    6. Green
    7. White-Brown
    8. Brown
4. Insert the arranged wires into the RJ45 connectors.
5. Crimp both connectors using the crimping tool.
6. Test the cable by connecting a computer to a switch.

### B) CROSSOVER CABLE

1. Take an Ethernet cable and strip the insulation from both ends.
2. Arrange the wires at one end according to the T568A standard.
3. Arrange the wires at the other end according to the T568B standard.
4. The T568A colour sequence is:
    1. White-Green
    2. Green
    3. White-Orange
    4. Blue
    5. White-Blue
    6. Orange

7. White-Brown
8. Brown

5. Insert the wires into the RJ45 connectors without changing their order.
6. Crimp both ends using the crimping tool.
7. Test the cable by connecting two computers directly.

## OUTPUT:

### Straight-Through Cable:

Both ends follow the same wiring standard (T568B – T568B).

The cable is prepared for connecting a computer to a switch or router.

### Crossover Cable:

One end follows T568A and the other follows T568B.

The cable is prepared for directly connecting two computers.

## RESULT:

Thus, the different types of network cables were studied, and straight-through and crossover Ethernet cables were prepared using RJ45 connectors and a crimping tool. The wiring arrangements and their applications in computer networks were understood.